Milestone Two Guidelines and Rubric

Module C: Data Science Capstone

Overview

In Milestone Two, you will use machine learning models to complete data analysis and evaluation of multiple datasets. Each week's content will cover a new topic or revisit topics from previous Mods. Using the project that you selected in Module B, you will apply each week's topic(s) and perform multiple analyses on your project's datasets. Your work should be reflected in your weekly Jupyter Notebook homework. Your project will be worked on individually, and your solutions will be submitted as an individual summary document in Week 12.

In this course, there are no specific questions to answer in your Jupyter Notebook files; your general goal is to analyze your data using the methods you have learned in this course and in this program, and to draw interesting conclusions based on the project you have chosen. In some cases, you may already have a clear question you want to answer based solely on what you see in your data set. In other cases, you may need to explore multiple methods and analyses before determining which questions should or need to be answered.

This is an iterative process; one that is designed to be self-governed and open-ended. As you complete each milestone, you are developing the narrative behind the data set and, by the time you are ready to complete your Integrated Capstone Project, determining the key questions that a client related to your project would be interested in having answered.

Requirements

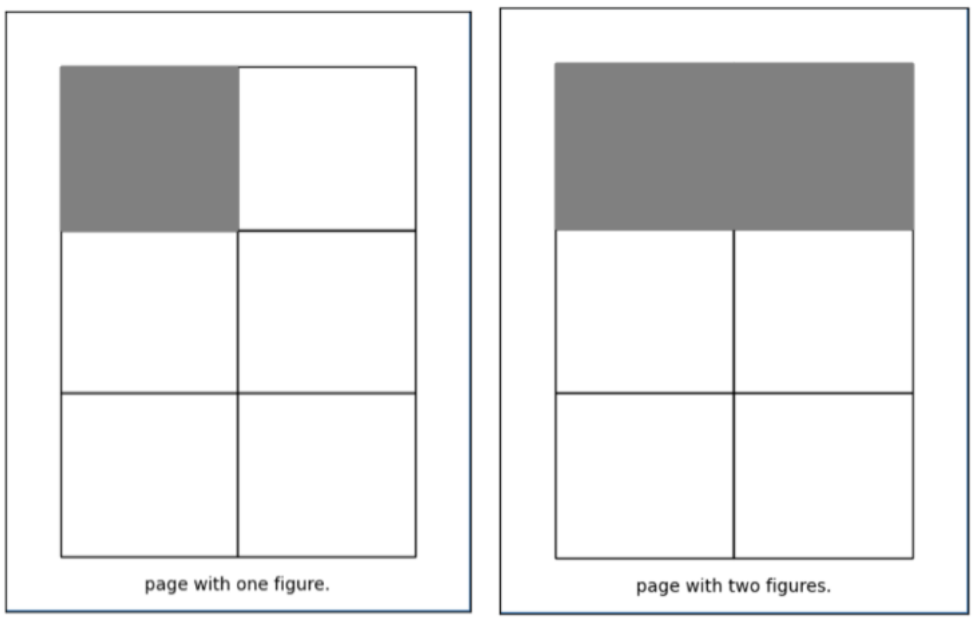
Criteria for Milestone One include:

1. A statement or description of the chosen project

2. An explanation of information learned from using the following modeling techniques. Furthermore, you must select one to two of these weeks (a–d) to discuss in a thorough and interesting way:
    - K-nearest neighbors (with different metrics) (Week 8)

- Gradient boost (Week 9)
- K-Means and silhouette score (Week 10)
- Density-based spatial clustering of applications with noise (DBSCAN) and hierarchical agglomerative clustering (Week 11)

3. Figures supporting your analysis (above)
    - You are expected to include 3-8 figures to support your conclusions about the data
    - A single page should fit
        - A **maximum** of 6 images per page
        - Image size should reference the A4 layout example below – the key is that the image content must be clearly visible.



- Figures should be included **in the body of your submission** and not as an appendix.

- Tables count towards your total number of figures
- Captions and figure titles **do not count** towards the dimensions of the figures. Captions should be no smaller than the minimum required font-size of 12pt Times New Roman
- Note that you may make figures larger than the specified size, but not smaller. The goal is to ensure that figures are clear, legible, and easy to read.

- For each topic above (2a–2d), an explanation of the following (as applicable to your dataset):
- How did you avoid overfitting? You should mention the techniques used to prevent overfitting, why these techniques were expected to be helpful, and the results.
- What metric(s) did you use, and what was the result? Did you use any hyperparameter tuning? You should describe the metrics and hyperparameter tuning performed to determine hyperparameters and include a discussion of the meaning of these parameters.
- What aspects of the results were expected or unexpected?
- How did your Exploratory Data Analysis help with the modeling?
- What sources did you rely on, apart from the course materials, to learn about this model. In this context, it is expected that you will find external sources, such as free educational URLs, and share and discuss them with other students through YellowDig.

4. Specific and supported conclusions drawn from the models.
    - How has your work up to this point helped address any research questions or other questions of interest?
    - Provide an explanation as to how you know the conclusions are true, potentially addressing any or all of the topics mentioned in Requirement 3 (i.e., overfitting, metrics, etc.)
    - This explanation should show that:
        - The conclusions are true from a quantitative perspective, and
        - Explain why the conclusions are relevant and/or important.

In each case, it is up to you to choose and/or emphasize the items that are most interesting to talk about, but you must strive for both *breadth* and *depth.* That is, your grade will be based on:

- Breadth: Briefly discussing all of the topics in Requirement 2, and
- Depth: Choosing and discussing 1–2 weeks' topics in a very thorough and interesting way. Make sure you explicitly state in your document which topics you are delving deeply into explaining.

Formatting

- Your summary document must be an 8- to 10-page Microsoft Word document with double spacing, 12-point Times New Roman font, and one-inch margins.
- Your document must contain an appendix consisting of your Jupyter Notebook completions from week 8-12 of the course, showing how you applied the topics of each week to your project. A full link to the GitHub repo of your notebooks in the appendix is preferred.
    - There is no page limit for the appendix containing your Jupyter Notebooks and they are not included in the page requirement of this assignment.
    - It is expected that some material in the summary, such as graphs, could be the same as those from your Jupyter notebooks. Material taken from these files provides the evidence for your claims in the summary.
- Citations/Bibliography
    - Citations may follow APA, Chicago, or any other style that makes it easy for the grader to locate the source.
    - Minor differences in citation style (for example, using a slightly older version of APA) will **not** result in a deduction of points. The primary purpose of the citation requirement is to ensure that the grader can easily locate the sources you are referencing, if needed.

Milestone Two Rubric

You will receive a grade from the LF based on where your assignment falls in the following scale:

- Exemplary: Meets all or nearly all expectations. 100% of points in the criteria
- Proficient: While meeting most expectations, the submission is lacking some details or insights. It is otherwise mostly correct and well-written. 85% of points in the criteria
- Needs improvement: While meeting many expectations and being correct in many ways, the submission is overly brief, incomplete, superficial, inaccurate, unclear and/or minimal in some ways. 70% of points in the criteria
- Not evident: Does not discuss and/or incorrect. 0% of points in the criteria

The table below shows what you need to do to achieve an "Exemplary" for each category.

Criteria	Exemplary	Points
Problem Statement/ Description	Provides a clear and detailed description of the chosen project, including insight into the potential impact of the project.	5
Conclusions: Depth	Provides accurate and insightful descriptions of how the chosen week(s)' modeling techniques were used on the project datasets, including evidence-supported conclusions that were derived from them and how the issues of those models (overfitting, etc.) were mitigated.	30
Conclusions: Breadth	Provides clear and detailed information about all modeling techniques covered in Weeks 8–11, including evidence-supported conclusions that were derived from them and accurately describing the issues of those models (overfitting, etc.)	30
Overfitting	Provides a comprehensive and insightful explanation of overfitting including techniques to prevent overfitting, how these techniques were expected to help, and the results.	10

Metrics and Hyperparameter Tuning	Provides a comprehensive and insightful explanation of how metrics and hyperparameter tuning were performed and the meaning of the hyperparameters as it connects to the problem.	5
Expected/ Unexpected	Provides detailed and insightful discussion of expected/unexpected aspects of data as evidenced by models.	5
Exploratory Data Analysis (EDA)	Provides detailed and insightful discussion of the usefulness/non-usefulness of EDA for model results.	5
AI Appendix and Citations	Includes an AI appendix clearly stating use of AI in submission. Citations are included and an approved citation style is used.	5
Communication	Consistently and effectively communicates ideas clearly, concisely, and effectively for a specific audience.	5
Total		**100**